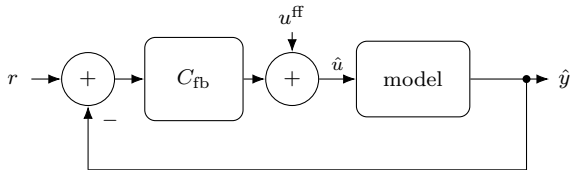


$$u = C_{fb}(r - y) + u^{ff}$$

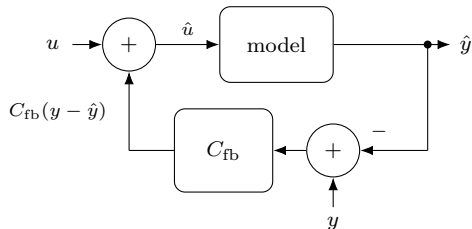
recorded



$$\hat{u} = C_{fb}(r - \hat{y}) + u^{ff}$$

what we want

$r - \hat{y} = (r - y) + (y - \hat{y})$, and C_{fb} is linear,
so $C_{fb}(r - \hat{y}) + u^{ff}$ becomes $u + C_{fb}(y - \hat{y})$.



$$\hat{u} = u + C_{fb}(y - \hat{y})$$

what we run

The correction $C_{fb}(y - \hat{y})$ is the force the controller would have added because of the model's error.

It is zero when $\hat{y} = y$, i.e. a perfect model is fed exactly the recorded u .

y and $\hat{y}$ enter only as numbers, so the step above needs C_{fb} linear and nothing else. Rows 1 and 2 are the same drawing with the block swapped; the plant and the model are never opened.